



6356 - A Bonanza of Dusty, Old Supernovae in Archival Observations

Cycle: 3, Proposal Category: AR

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OBSERVATIONS

ABSTRACT

While transient astronomy is often associated with young, rapidly evolving sources, there is a subset of dusty, old supernovae (SNe) that are quite valuable to our understanding of SN explosions and their progenitor systems. The near- and mid-IR wavelengths span the peak of the thermal emission from such dust and are best to characterize the dust in these systems. Yet the phase space of existing observations (in terms of dust temperature and SN age) remains relatively unpopulated owing to the combined low sensitivity and/or limited IR wavelength coverage of existing telescopes. JWST offers a unique opportunity to detect a large sample of these SNe at long wavelengths and late times. With literally thousands of SNe having exploded in nearby (<200 Mpc) galaxies over the past 50-100 years, an archival study stands to yield a bonanza of low-hanging fruit. Here we propose an Archival Research (AR) program to search for dusty, surviving targets in JWST observations of locations of over 200 known SNe that were/will be observed serendipitously in Cycles 1 & 2. Although relying on archival data may not always result in a complete spectral energy distribution (SED) for each SN, it is a low-risk, easy, and cheap (time-wise) method for taking advantage of existing data to build an initial sample, which can ultimately be used to enable a variety of new and important types of scientific investigations, and set a foundation for additional follow-up and monitoring in future Cycles.

OBSERVING DESCRIPTION

An AR proposal to search archival data at the positions of roughly 300 SNe for old, dusty SNe.